

SCALABILITY AND RESILIENCE IN DISTRIBUTED ENVIRONMENTS.

Next-Gen Microservices Architecture

CURRENT ARCHITECTURE

Monolith Bottlenecks

Current monolithic core is tightly coupled. Deployment cycles take 4 hours. Database locks frequent during peak event traffic.

NEW TOPOLOGY

Event-Driven Microservices

Transitioning to Kafka for asynchronous messaging.
Services containerized via Kubernetes. API Gateway
handling routing and rate limiting.

SECURITY & COMPLIANCE

Zero-Trust Protocol

Implementing strictly authenticated inter-service communication via mutual TLS. Regular automated vulnerability scanning in the CI/CD pipeline.

MIGRATION PLAN

Rollout Phases

Phase 1: Decoupling the Authentication module. Phase 2: Migrating the Data warehouse layer. Phase 3: Sunset of legacy monolith.